Timothy Do

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Education

University of California, Los Angeles — Sep. 2023 – March 2025 *(Expected)*
M.S. Electrical and Computer Engineering, Signal Processing Track — *Los Angeles, CA*
- **Relevant Coursework:** Matrix Analysis, Computational Imaging, Embedded Systems

University of California, Irvine — Sep. 2019 – June 2023
B.S. Electrical Engineering, Accelerated Status — *Irvine, CA*
- **Cumulative GPA:** 3.924/4.0, **Major GPA:** 3.944/4.0, **Magna Cum Laude**
- **Specializations:** Digital Signal Processing, Communications
- **Relevant Coursework:** Computer Architecture, Control Systems, Digital Image/Signal Processing, Electronics
- **Clubs and Societies:** Irvine Computer Vision Laboratory, IEEE, IPC, Eta Kappa Nu, Tau Beta Pi

Experience

Embedded Software Engineering Intern — June 2023 –
Maxeon Solar Technologies — *San Jose, CA*
- Implemented Robust OpenCV pipelines to detect contours/lines with multiprocessing acceleration
- Developed Tkinter GUI interfaces for operator ease of use when applying image pipelines to test images
- Performed Multi-Class Data Labeling and Implemented Data Augmentation methods for training a robust YOLOv5 Image Defect Classification Model

Industrial Internet of Things Intern — June 2022 – Sep. 2022
Western Digital — *San Jose, CA*
- Applied Industrial Toolsets to develop a Raspberry Pi Sensor Data Analytics Platform
- Earned the Six Sigma Yellow Belt for learning foundations of Lean Six Sigma
- Saved the company an annual projected cost of $312,000 from hazard mitigations

Undergraduate Student Researcher — January 2022 – Present
Irvine Computer Vision Laboratory at UC Irvine — *Irvine, CA*
- Researched under Professor Glenn Healey on the topic of Spectral Image Filtering and Baseball Simulations.
- Digitized Hyperspectral Imaging Filters into CSVs and compared them against certain chemical emissions.
- Led the lab in analyzing Hyperspectral Data Cubes and determined dominant bands in SWIR and NWIR spectras

Projects

The Do-Pro Senior Design Project | *Raspberry Pi, Python, MATLAB, OpenCV* — Sep. 2022 - March 2023
- Led a team of four in creating an embedded stereo-vision camera for ADAS detection and 3D-Reconstruction
- Calibrated the stereo-camera in MATLAB and implemented block matching algorithms with OpenCV
- Designed the button/touch-screen user interface using RPi.GPIO and Tkinter for navigating the camera
- Developed a Power Regression Model for the Do-Pro to percieve depth with 6% error from 1-3 feet

AM Radio & Amplifier | *LTSpice, Oscilloscopes, Soldering* — April 2022 – May 2022
- Led IPC @ UCI's Two-Part AM Radio Workshop, Creating the Schematic for Optimal Gain and Efficiency
- Designed Audio Amplifier using the *LM741* Op-Amp in an Inverting Configuration with a Single USB 5V Supply
- Designed Envelope Detector using the *1N4148* Diode and RC Band-Pass Filter for Demodulation

EECS195-Colorization | *Python, MATLAB, OpenCV, Tensorflow* — Feb. 2022 – March 2022
- Colorized grayscale images using GAN and Auto-Encoder Neural Networks
- Gathered own datasets by taking pictures of different scenery
- Clustered pixels into superpixel subgraphs for uniform training weights

Skills

Electronics: Raspberry Pi, Network Analysis, Cadence, LTSpice, Soldering, Oscilloscope & Multimeter Measurements
Programming Languages: Python, Java, C, CUDA, MATLAB, bash, MySQL HTML/CSS/Javascript, VHDL
Developer Tools: git, SSH, Makefile, Linux, Azure, VS Code, Visual Studio, PyCharm, Eclipse, Vivado HLS
Libraries: OpenCV, Pandas, Numpy, Matplotlib, PIL, Tensorflow, Pytorch, Tkinter RPi.GPIO, Socket, Flask